

Tanner J. Berman

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Education

University of California San Diego

Bachelor of Science in Computer Science

GPA: 3.73

Expected Graduation: Fall 2026

Programming Languages & Tools: C++, C, Go, Python, HTML/CSS/JavaScript, React, PostgreSQL, MongoDB, Express, n8n, TypeScript, Docker

Relevant Coursework: Operating Systems, Computer Networks, Parallel Programming, Data Structures & Algorithms, Computer Architecture, Wireless Embedded Systems, Software Engineering, Linear Algebra, Statistical Methods

Experience

Parsley360, Software Engineer

San Diego, CA

September 2025 - Present

- Engineered an automated ETL pipeline utilizing n8n and Apify, writing custom JavaScript modules to parse, transform, and sanitize highly unstructured data before routing the payloads into a PostgreSQL database.
- Solved critical network and memory bottlenecks for massive binary payloads by implementing custom execution loops and chunked batching, entirely eliminating out-of-memory crashes and ECONNRESET timeouts.
- Designed a fault-tolerant Supabase PostgreSQL schema leveraging advanced SQL and JSONB parsing to manage complex queue states, data deduplication, and relational cloud storage mapping.
- Optimized data extraction reliability by reverse-engineering network request patterns and implementing dynamic HTTP header generation to bypass Cloudflare WAF.

Parsley360, SWE Intern

San Diego, CA

June 2025 - September 2025

- Built a scalable, multimodal data-ingestion pipeline (Python + C++) that scrapes, OCRs, and cleans documents, cutting manual preprocessing time by ~90%.
- Designed and implemented a DuckDB/MotherDuck (SQL-based) corpus database with URI links to S3 storage, SHA-256 dedup; delivered clean topics/embeddings for the demo's reasoning engine.
- Applied Gensim LDA and BERTopic for topic-driven RAG and ontology mapping; wrote a C++ tokenizer.

Software Lead and Event Host at ACM UCSD

San Diego, CA

January 2025 - June 2025

- Led a 6-person Agile team to architect a MERN stack web app matching UCSD students for rides to airports.
- Built MongoDB schemas via Mongoose/Atlas, integrated Firebase auth, and implemented post filtering.
- Facilitated Agile ceremonies, mentored junior developers, and oversaw backend PR reviews.
- Led and organized multiple events, including a well-received intern panel with 50+ attendees.

Projects

Reverse-Engineered Smart-Bulb BLE Remote (C++, ESP32, BLE, Cryptography)

July 2026 - Present

- Reverse-engineered the undocumented BLE protocol of AiDot/Leedarson smart bulbs (APK decompilation via jadx, Wireshark, cryptanalysis), reconstructing its SHA-256 key derivation and AES-256-CTR command cipher.
- Built ESP32 firmware in C++ (NimBLE, mbedTLS) performing the auth handshake to control multiple bulbs locally, no app, cloud, or Wi-Fi dependency.
- Discovered and responsibly documented a cryptographic weakness (per-session IV/keystream reuse) enabling keyless decryption and forgery of commands.

Broccoli Game Platform (Go, WebSockets, JavaScript)

November 2025 - Present

- Architected a local-authoritative multiplayer game engine in Go, utilizing WebSockets to handle real-time state synchronization between a desktop host and mobile clients.
- Engineered a seamless deployment pipeline leveraging Ngrok to establish secure TCP cloud tunnels, bypassing NAT restrictions and allowing remote players to join instantly via browser without port-forwarding.
- Packaged cross-compiled binary executables for macOS and Windows, abstracting local server initialization and networking into single-click launch scripts.

Low-Power BLE Tracker (C, STM32L4, I2C)

January 2025 - April 2025

- Pair-programmed firmware for a BLE-based tracker on STM32L4 hardware, designed to detect when an item is left behind and broadcast a lost signal via BLE.
- Implemented custom drivers for I2C sensors, timers, accelerometer (LSM6DSL), and BLE communication.
- Optimized system power consumption from 110 mA to ~80 μ A using STOP2 mode, LSI-driven timers, and aggressive peripheral gating for multi-month battery life.